Every Algebraic Number is a Critical Value of an Integer Polynomial

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Abstract

For every nonconstant $f \in \mathbb{Z}[X]$ we construct $g \in \mathbb{Z}[X]$ and an irreducible $H \in \mathbb{Z}[X]$ with $H \mid g'$ and $H^2 \mid f \circ g$. Equivalently, every algebraic number is a critical value of an integer polynomial, so the set of critical values of $\mathbb{Z}[X]$ is exactly $\overline{\mathbb{Q}}$. The construction is explicit. The proof was assisted by AI, and has been formalized in Lean.

1 Introduction

A critical value of a nonconstant $g \in \mathbb{Z}[X]$ is a number α with $g(\beta) = \alpha$ and $g'(\beta) = 0$ for some β. Since $\overline{\mathbb{Q}}$ is closed under polynomial evaluation, every critical value is algebraic. This note proves the converse, and does so with an explicit construction.

Theorem 1. *Let $f \in \mathbb{Z}[X]$ be nonconstant. Then there exist $g, H \in \mathbb{Z}[X]$ with H irreducible in $\mathbb{Z}[X]$ and $1 \leq \deg H$, such that*

$$H \mid g' \quad \textit{and} \quad H^2 \mid f \circ g \qquad \textit{in } \mathbb{Z}[X].$$

The g produced satisfies $\deg g \geq 2$.

The corollary is the statement in the title.

Corollary 1. *The set of critical values is exactly $\overline{\mathbb{Q}}$.*

2 The construction

Let $h \in \mathbb{Z}[X]$ be a primitive irreducible factor of f, of degree n, with root θ. If $h = X$, take $g = X^2$ and $H = X$; then $H \mid g'$ and $H^2 \mid f \circ g$. Otherwise $h(0) \neq 0$, and since $\gcd(h, h') = 1$ in $\mathbb{Q}[X]$, the elimination ideal $(h, h') \cap \mathbb{Z}$ is a nonzero ideal of $\mathbb{Z}$; let ρ generate it,

$$|\rho| = \min\{\, |c| \;:\; c \in (h, h') \cap \mathbb{Z},\ c \neq 0 \,\}, \qquad u\,h + v\,h' = \rho \quad (u, v \in \mathbb{Z}[X]),$$

which the extended Euclidean algorithm in $\mathbb{Q}[X]$ produces on clearing denominators, and set

$$M = \rho\, h(0), \qquad H(X) = \frac{h(MX)}{h(0)}, \qquad W(X) = \frac{v(0)H(X) - v(MX)}{\rho}.$$

The polynomial claimed by Theorem 1 is

$$\boxed{g(X) = MX + h(MX)\, W(X)}$$

together with this H.

Theorem (restatement of Theorem 1)**.** *H is irreducible in $\mathbb{Z}[X]$, and $H \mid g'$ and $H^2 \mid f \circ g$ in $\mathbb{Z}[X]$.*

Verification. Let $\beta = \theta/M$, a root of H. Since $H(0) = 1$, the constant terms in W cancel, and every other coefficient carries M^j with $\rho\, h(0) \mid M$, so $W \in \mathbb{Z}[X]$ and $W(0) = 0$; likewise $H, g \in \mathbb{Z}[X]$. Then $W(\beta) = -v(\theta)/\rho = -1/h'(\theta)$, so

$$g(\beta) = \theta, \qquad g'(\beta) = M\big(1 + h'(\theta) W(\beta)\big) = 0.$$

Hence β is a double root of $f \circ g$, and H is primitive irreducible, so $H \mid g'$ and $H^2 \mid f(g)$ in $\mathbb{Z}[X]$ by Gauss.

The computation uses $h(\theta) = 0$ and nothing else about θ, so it runs at every root of h at once: all n conjugates are critical values of the same g, at the n points θ_i/M. That is what makes H^2, and not merely $(X-\beta)^2$, divide $f \circ g$.

Only the relation $uh + vh' = \rho$ is used, so any nonzero element of $(h, h') \cap \mathbb{Z}$ serves — $\mathrm{Res}(h, h')$ among them, which is a multiple of ρ. The generator is taken because $M = \rho\, h(0)$ multiplies into every coefficient of H, W and g, so the smallest ρ gives the smallest answer. On $h = X^2 + 1$ the resultant is 4 against $\rho = 2$, and on Dedekind's cubic below it is 2012 against 1006.

3 Examples

Each line below is a closed identity, checkable by expansion.

Example 1 ($f = qX - p$, $p \neq 0$, $\gcd(p, q) = 1$). Here $h' = q$ and $p = qX - h$, so $(h, h') \cap \mathbb{Z} = \mathbb{Z}$ and $\rho = 1$. Choose $a, b \in \mathbb{Z}$ with $ap + bq = 1$ and take $u = -a$, $v = aX + b$; then $h_0 = -p$, $M = -p$, $H = qX + 1$, $W = X$, and

$$g = -pqX^2 - 2pX, \qquad g' = -2p\,(qX+1), \qquad f \circ g = -p\,(qX+1)^2.$$

Note $\deg v = n$: the minimal ρ is not in general attainable with $\deg v < n$, which for this h would force $q \mid \rho$.

Example 2 ($f = X^2 + 1$). With $u = 2$, $v = -X$, $\rho = 2$, $h_0 = 1$, $M = 2$, $H = 4X^2 + 1$ and $W = X$,

$$g = 4X^3 + 3X, \qquad g' = 3(4X^2+1), \qquad f \circ g = (4X^2+1)^2(X^2+1).$$

Here $\deg g = 3 = n + 1$, the least degree any solution can have. The unit group $\mathcal{O}_K^\times = \{\pm 1, \pm i\}$ has rank 0, and it is $\mathbb{Z}[i/2] = \mathbb{Z}[i][1/2]$ that supplies the needed unit $1 + i$.

Example 3 ($f = X^3 - 2$). With $u = 3$, $v = -X$, $\rho = -6$, $h_0 = -2$, $M = 12$, $H = 1 - 864X^3$ and $W = -2X$,

$$g = -3456X^4 + 16X, \qquad g' = 16\,(1 - 864X^3).$$

Example 4 ($f = X^3 - X - 1$). The construction gives $M = 23$ and a sextic. A hand-tuned quintic also witnesses the *statement*, and is a useful independent check of it:

$$g = -24334X^5 + 39675X^4 - 26450X^3 + 9085X^2 - 1610X + 118, \qquad H = 23X^3 - 23X^2 + 8X - 1,$$

with $g' = -230\,(23X - 7)H$ and $H^2 \mid f \circ g$, the cofactor having degree 9.

Example 5 (Dedekind's cubic). $f = X^3 - X^2 - 2X - 8$, the standard example of a field whose ring of integers is not monogenic: 2 is a common index divisor [Ded78]. With $u = 21X - 125$, $v = -7X^2 + 44X - 3$, $\rho = 1006$, $h_0 = -8$, $M = -8048$,

$$H = 65158925824X^3 + 8096288X^2 - 2012X + 1, \qquad W = -194310912X^3 + 426544X^2 + 358X,$$

$$\begin{aligned} g = {} & 1012887224144143319 04X^6 - 2097596140126208 00X^5 \\ & - 217370176548864X^4 - 14767629312X^3 + 2350016X^2 - 10912X. \end{aligned}$$

4 Formalization

The development above has been formalized in Lean 4 [MU21] against Mathlib [The20], in about 1,200 lines. The main statement is

```
theorem main (f : Z[X]) (hf : 1 <= f.natDegree) :
    exists g H : Z[X], Irreducible H /\ H | derivative g /\ H^2 | f.comp g
```

with a strengthened form carrying $2 \leq \deg g$ and $1 \leq \deg H$, and with Corollary 1 proved from it. Every result depends only on the three standard axioms `propext`, `Classical.choice` and `Quot.sound`.

References

[Ded78] Richard Dedekind. “Über den Zusammenhang zwischen der Theorie der Ideale und der Theorie der höheren Kongruenzen”. In: *Abhandlungen der Königlichen Gesellschaft der Wissenschaften zu Göttingen* (1878).

[The20] The mathlib Community. “The Lean Mathematical Library”. In: *Certified Programs and Proofs (CPP 2020)*. ACM, 2020.

[MU21] Leonardo de Moura and Sebastian Ullrich. “The Lean 4 Theorem Prover and Programming Language”. In: *Automated Deduction – CADE 28*. Lecture Notes in Computer Science. Springer, 2021.